Research Notes on Stochastic gradient descent

Stochastic gradient descent

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta L(\theta_t)$$

>> Section 1

for Σ (w) = 1 n 2 (∑ i = 1 n Q i (w) − Q (w)) (∑ i = 1 n Q i (w) − Q (w)) T {\displaystyle \Sigma (w)={\frac {1}{n^{2}}}\left(\sum _{i=1}^{n}Q_{i}(w)-Q(w)\right)\left(\sum _{i=1}^{n}Q_{i}(w)-Q(w)\right)^{T}} where d B t {\textstyle dB_{t}} denotes the Ito-integral with respect to a Brownian motion is a more precise approximation in the sense that there exists a constant C > 0 {\textstyle C>0} such that

>> Section 2

Nesterov-enhanced gradients: NAdam, FASFA varying interpretations of second-order information: Powerpropagation and AdaSqrt. Using infinity norm: AdaMax AMSGrad, which improves convergence over Adam by using maximum of past squared gradients instead of the exponential average. AdamX further improves convergence over AMSGrad. AdamW, which improves the weight decay.

>> Section 3

w := w − η ∇ Q (w) = w − η n ∑ i = 1 n ∇ Q i (w) . {\displaystyle w:=w-\eta \,\nabla Q(w)=w-{\frac {\eta }{n}}\sum _{i=1}^{n}\nabla Q_{i}(w).}

>> Section 4

Further proposals include the momentum method or the heavy ball method, which in ML context appeared in Rumelhart, Hinton and Williams' paper on backpropagation learning and borrowed the idea from Soviet mathematician Boris Polyak's 1964 article on solving functional equations. Stochastic gradient descent with momentum remembers the update Δw at each iteration, and determines the next update as a linear combination of the gradient and the previous update:

>> Section 5

RMSProp RMSProp (for Root Mean Square Propagation) is a method invented in 2012 by James Martens and Ilya Sutskever, at the time both PhD students in Geoffrey Hinton's group, in which the learning rate is, like in Adagrad, adapted for each of the parameters. The idea is to divide the learning rate for a weight by a running average of the magnitudes of recent gradients for that weight. Unusually, it was not published in an article but merely described in a Coursera lecture.

>> Section 6

G j , j = ∑ τ = 1 t g τ , j 2 . {\displaystyle G_{j,j}=\sum _{\tau =1}^{t}g_{\tau ,j}^{2}.} This vector essentially stores a historical sum of gradient squares by dimension and is updated after every iteration. The formula for an update is now

>> Section 7

Q (w) = ∑ i = 1 n Q i (w) = ∑ i = 1 n (y ^ i − y i) 2 = ∑ i = 1 n (w 1 + w 2 x i − y i) 2 . {\displaystyle Q(w)=\sum _{i=1}^{n}Q_{i}(w)=\sum _{i=1}^{n}\left({\hat {y}}_{i}-y_{i}\right)^{2}=\sum _{i=1}^{n}\left(w_{1}+w_{2}x_{i}-y_{i}\right)^{2}.}

>> Section 8

Linear regression Suppose we want to fit a straight line y ^ = w 1 + w 2 x {\displaystyle {\hat {y}}=w_{1}+w_{2}x} to a training set with observations ((x 1 , y 1) , (x 2 , y 2) … , (x n , y n)) {\displaystyle ((x_{1},y_{1}),(x_{2},y_{2})\ldots ,(x_{n},y_{n}))} and corresponding estimated responses (y ^ 1 , y ^ 2 , … , y ^ n) {\displaystyle ({\hat {y}}_{1},{\hat {y}}_{2},\ldots ,{\hat {y}}_{n})} using least squares. The objective function to be minimized is

>> Section 9

Sign-based stochastic gradient descent Even though sign-based optimization goes back to the aforementioned Rprop, in 2018 researchers tried to simplify Adam by removing the magnitude of the stochastic gradient from being taken into account and only considering its sign. This results in a significantly lower communication cost of transferring gradients from workers to the parameter server. In this sense, it serves to better compress the gradient information, while having comparable convergence to standard SGD.

>> Section 10

Implicit updates (ISGD) As mentioned earlier, classical stochastic gradient descent is generally sensitive to learning rate η. Fast convergence requires large learning rates but this may induce numerical instability. The problem can be largely solved by considering implicit updates whereby the stochastic gradient is evaluated at the next iterate rather than the current one:

>> Section 11

The scaling factor ξ ∗ ∈ R {\displaystyle \xi ^{\ast }\in \mathbb {R} } can be found through the bisection method since in most regular models, such as the aforementioned generalized linear models, function q ( ) {\displaystyle q()} is decreasing, and thus the search bounds for ξ ∗ {\displaystyle \xi ^{\ast }} are [min (0 , f (0)) , max (0 , f (0))] {\displaystyle [\min(0,f(0)),\max(0,f(0))]} .

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>> Section 12

Notable applications Stochastic gradient descent is a popular algorithm for training a wide range of models in machine learning, including (linear) support vector machines, logistic regression (see, e.g., Vowpal Wabbit) and graphical models. When combined with the backpropagation algorithm, it is the de facto standard algorithm for training artificial neural networks. Its use has been also reported in the Geophysics community, specifically to applications of Full Waveform Inversion (FWI). Stochastic gradient descent competes with the L-BFGS algorithm, which is also widely used. Stochastic gradient descent has been used since at least 1960 for training linear regression models, originally under the name ADALINE. Another stochastic gradient descent algorithm is the least mean squares (LMS) adaptive filter.

>> Section 13

Second-order methods A stochastic analogue of the standard (deterministic) Newton–Raphson algorithm (a "second-order" method) provides an asymptotically optimal or near-optimal form of iterative optimization in the setting of stochastic approximation. A method that uses direct measurements of the Hessian matrices of the summands in the empirical risk function was developed by Byrd, Hansen, Nocedal, and Singer. However, directly determining the required Hessian matrices for optimization may not be possible in practice. Practical and theoretically sound methods for second-order versions of SGD that do not require direct Hessian information are given by Spall and others. (A less efficient method based on finite differences, instead of simultaneous perturbations, is given by Ruppert.) Another approach to the approximation Hessian matrix is replacing it with the Fisher information matrix, which transforms usual gradient to natural. These methods not requiring direct Hessian information are based on either values of the summands in the above empirical risk function or values of the gradients of the summands (i.e., the SGD inputs). In particular, second-order optimality is asymptotically achievable without direct calculation of the Hessian matrices of the summands in the empirical risk function. When the objective is a nonlinear least-squares loss